

Evidence-Bounded Visual Multimodal Privacy Protection for Egocentric Personal Images

Literature Review

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Abstract

This literature review synthesises 34 papers (one CASTLE 2024 anchor and 33 papers across four themes: egocentric face detection, generative face anonymisation, multimodal screen and text privacy, and privacy evaluation including adversarial re-identification and utility metrics), motivated by building a privacy-preserving pipeline for CASTLE 2024, used here as an offline-extracted WebP still-image corpus comprising 416,542 frames across 16 streams and four recording days, with 11 unique participant identifiers and 10 active participants per day. The raw 8.22 TB video collection is not used in this practicum.

Across themes, the field advances individual sub-problems while offering limited evidence of integrated performance under realistic egocentric conditions. Many generative anonymisation methods achieve strong results on controlled portrait datasets yet are seldom evaluated under motion blur, non-frontal viewpoints, or the lossy compression of wearable-camera capture. Privacy evaluation predominantly relies on face-level re-identification, while contextual and demographic channels (which can reach 65–80% attack accuracy without face access) receive inconsistent treatment. Screen and text privacy and face anonymisation are not evaluated jointly within the reviewed corpus, and utility metric reporting remains inconsistent across methods.

Three gaps are formalised: the absence of quality-aware adaptive strategy selection; limited cross-view identity-consistency evaluation in multi-stream settings using filename-derived temporal alignment; and the absence of a unified multimodal privacy pipeline validated on real egocentric lifelog data. These gaps motivate the four research questions and methodology that follow.

Index Terms

Egocentric vision, lifelog privacy, face anonymisation, multimodal privacy, generative models, re-identification, CASTLE dataset.

I. INTRODUCTION

A. Wearable Lifelogging and the Privacy Imperative

The past decade has seen wearable cameras transition from experimental laboratory devices into viable consumer technology, enabling continuous first-person recording (or lifelogging) across the full spectrum of daily activity. Devices such as GoPro action cameras and smart glasses can now capture uninterrupted egocentric video at 4K resolution and 50 frames per second, accumulating terabytes of footage per participant per session. The scientific value of this data class is considerable: Grauman et al. demonstrated its scope with the Ego4D corpus (3,670 hours of first-person video collected from 931 participants across 74 worldwide locations) [1], establishing egocentric vision as a distinct and productive research discipline encompassing action recognition, object interaction, and social scene understanding.

Yet the same properties that make lifelogging scientifically rich render it a severe privacy liability. A single lifelog frame may simultaneously capture the biometric identities of non-consenting bystanders, sensitive information visible on laptop screens, handwritten notes, and payment terminals (a compound, multimodal disclosure event that no existing privacy protection system addresses in full). The European Union’s General Data Protection Regulation classifies biometric data used for identification as a special category of personal data requiring explicit legal basis for processing (GDPR 2016/679, Article 9). Under this framework, distributing unanonymised lifelog footage is not merely ethically problematic but potentially unlawful. The tension between lifelogging’s scientific promise and its GDPR obligation constitutes the central problem addressed by this review.

B. The Egocentric Privacy Challenge: Beyond Face Anonymisation

Privacy challenges in egocentric footage are qualitatively distinct from those in conventional surveillance or portrait photography. Persons appear non-frontally, at variable distances, in motion, and under uncontrolled illumination (conditions that produce motion blur, extreme viewing angles, partial occlusion, and compression artefacts that standard face anonymisation pipelines, optimised for aligned studio-quality images, were not designed to handle). Khan et al. comprehensively survey de-identification methods across a decade of literature and confirm that robust evaluation under real-world degraded imaging conditions remains the exception rather than the rule [2]; methods that do claim robustness are assessed on controlled datasets such as CelebA-HQ, FFHQ, or LFW (none of which approximate the motion blur, resolution variability, or non-frontal angles of egocentric capture).

Beyond face identity, egocentric data exposes a broader spectrum of privacy-sensitive content that face anonymisation alone cannot suppress. Visible laptop screens, digital displays, handwritten notes, and printed documents constitute screen content and text PII that are systematically present in workplace and daily-life lifelogs. Li et al. provide an empirical lower bound on this broader threat: without any face detection whatsoever, zero-shot foundation models recover demographic attributes from egocentric video at peak accuracies of 73.15% (gender), 79.64% (age), and 65.36% (race) across models (Tab. 2 of [3]) (each substantially above random-chance performance). This finding establishes that a comprehensive privacy framework must address face identity, screen content, and text obfuscation as an integrated multimodal problem, not as isolated sub-tasks.

C. CASTLE 2024: The Evaluation Anchor

The CASTLE 2024 dataset provides the anchoring evaluation context for this review [4]. Captured using GoPro HERO10 Black cameras at $3,840 \times 2,160$ UHD 4K resolution and 50 fps, the original CASTLE release comprises 8.22 TB of multi-view lifelog footage. This practicum does not process the raw video collection. Instead, it accesses the offline-extracted WebP still-frame corpus derived from CASTLE 2024: 416,542 frames across 16 streams (10 egocentric streams, 5 fixed exocentric streams, and 1 additional egocentric stream arising from a Day 4 participant substitution), 11 unique participant identifiers, 10 active participants per day, and four recording days. The accessible project corpus contains no separate per-frame timestamp file or cross-stream synchronisation metadata beyond the filename-derived alignment mechanism considered later in the review; evaluation is image-based throughout. Faces and screen content are deliberately unblurred, placing a direct obligation on any derived publication under the dataset’s CC BY-NC-SA 4.0 licence [4]. CASTLE’s closed per-day 10-active-participant identity pool, drawn from 11 unique participant identifiers across the corpus, distinguishes it from open-world egocentric datasets: since the active identity set is bounded, it can support closed-set re-identification evaluation that open-world corpora such as Ego4D cannot support as directly.

D. Survey Question and Research Questions

Survey Question: “What is the state of the art in automated, adaptive, multimodal privacy protection (encompassing face anonymisation, screen content redaction, and text obfuscation) for high-resolution

egocentric lifelog imagery, with respect to privacy-utility trade-offs under real-world capture conditions and constrained compute environments?”

This question directly structures the four sections of this review and four corresponding research questions that define the practicum’s scope. RQ1 asks Which face detection architectures achieve the highest robustness on 4K egocentric frames exhibiting motion blur, partial occlusion, and extreme viewing angles under practical on-device compute constraints? RQ2 asks how GAN-based and diffusion-based face anonymisation methods compare on the privacy-utility trade-off (specifically re-identification resistance versus perceptual quality preservation) when applied to egocentric imagery. RQ3 asks what integrated pipeline architecture can simultaneously detect and redact faces, on-screen content, and visible text in egocentric frames, and where current multimodal approaches fall short. RQ4 asks what adaptive selection mechanism, conditioned on per-frame image quality and content characteristics, can optimise the trade-off across the heterogeneous conditions present in CASTLE. RQ4 is the practicum’s primary proposed contribution; the evidence base motivating and constraining it is built across Sections II–V. All four RQs are fully stated in Table II.

E. Scope, Search Strategy, and Exclusions

This is a narrative thematic review (the review type confirmed as appropriate for the MCM practicum scope and timeline). Literature is drawn from 34 peer-reviewed papers identified through systematic search of IEEE Xplore, ACM Digital Library, CVF Open Access, and Web of Science. The corpus is bounded by four headline criteria: full peer-reviewed publication at a CORE A*/A/B conference or Q1/Q2 journal (IC-01 to IC-03); primary publication window of 2020–2026, with pre-2020 foundational works retained only where they remain the established baseline in active use (specifically ArcFace [5], CRAFT [6], FID [7], LPIPS [8], and SSIM [9]) under a documented exception (IC-04); direct relevance to at least one of the four project themes (IC-06); and quantitative results on a named public dataset (IC-07). The CASTLE dataset paper is included under a supervisor-approved preprint exception (IC-09) [4]. Full inclusion and exclusion criteria (IC-01 to IC-10; EC-01 to EC-10) are listed in Table I.

Key exclusions include: non-peer-reviewed publications (EC-01); out-of-scope domains (audio/speech privacy, video-temporal-only models, and fixed-camera surveillance systems) (EC-04); qualitative-only evaluations (EC-05); and arXiv-only papers unless a confirmed peer-reviewed proceedings or journal version is available. Implementation tools (e.g., PaddleOCR, SCRFD, YOLO11) are excluded under EC-01 and IC-01 as non-peer-reviewed engineering artefacts and are addressed in the Methodology chapter.

F. Structure of This Review

The review is organised into four thematic sections, each concluding with a comparative synthesis table pre-planned to carry the cross-paper comparison load. Section II examines face detection in egocentric data, focusing on the performance gap between standard benchmark results and robustness under CASTLE’s 4K, motion-blurred, non-frontal conditions. Section III surveys generative face anonymisation across GAN-based, diffusion-based, and reversible approaches, synthesising a persistent metric inconsistency that prevents valid cross-method comparison from reported values alone. Section IV addresses multimodal privacy protection, reviewing screen content detection and scene text recognition as the missing components of a unified PII pipeline (a gap confirmed across all reviewed works). Section V reviews privacy evaluation frameworks and utility metrics [3], [5], [10], establishing that face-level re-identification is insufficient to characterise the full privacy leakage profile of egocentric lifelog data. Section VI synthesises the three gap claims derived from CASTLE’s specific constraints and presents the four refined research questions that motivate the adaptive anonymisation framework proposed in this practicum.

II. FACE DETECTION FOR EGOCENTRIC LIFELOG DATA

A. The Egocentric Detection Problem

Face detection is the necessary precondition for every downstream anonymisation operation in a lifelog privacy pipeline: a face that is not detected cannot be anonymised, and an undetected face constitutes an

TABLE I
INCLUSION AND EXCLUSION CRITERIA (ICEC)

ID	Criterion
<i>Inclusion Criteria</i>	
IC-01	Peer-reviewed full papers: conference, journal, specialist workshop at A*/A venue
IC-02	Q1 or Q2 per Scimago SJR (IEEE TPAMI, IJCV, TIP, CVIU, TIFS, Scientific Reports)
IC-03	CORE A*/A priority; CORE B permitted for niche sub-topics (PETS, ICDAR)
IC-04	2020–2026 primary; pre-2020 foundational works with documented justification
IC-05	English-language publications only
IC-06	Relevance to at least one of four themes: face detection, generative anonymisation, screen/text, and privacy evaluation including utility metrics
IC-07	Quantitative results on a named public or established dataset
IC-08	Full text via DCU Library, IEEE Xplore, ACM DL, CVF Open Access, or Web of Science
IC-09	CASTLE exception: Rossetto et al. (2025) unconditionally included (supervisor-approved)
IC-10	Reproducibility: prefer papers with public code/dataset where two papers cover same topic
<i>Exclusion Criteria</i>	
EC-01	Exclude ArXiv preprints, technical reports, blog posts, white papers, implementation tools
EC-02	Exclude Q3/Q4 journals; CORE C or unranked conferences
EC-03	Exclude pre-2020 unless named foundational baseline (ArcFace, FID, SSIM, LPIPS, CRAFT)
EC-04	Exclude out-of-scope: audio/speech privacy, video-temporal-only, fixed-camera CCTV, medical imaging
EC-05	Exclude qualitative-only; proprietary-dataset-only; opinion/position papers
EC-06	Exclude if full text not accessible via DCU/IEEE/ACM/CVF/WoS
EC-07	Exclude all non-English publications
EC-08	Keep most recent/comprehensive version if duplicate; log excluded duplicate with reason
EC-09	No research group contributes >3 papers unless CASTLE-specific work requires it
EC-10	Must access and cite primary source directly (no secondary summary citations)

TABLE II
RESEARCH QUESTIONS AND LITERATURE REVIEW MAPPING

RQ	Type	Question	LR Section
RQ1		Which face detection architectures achieve the highest accuracy and robustness on high-resolution egocentric frames exhibiting motion blur, partial occlusion, and extreme viewing angles, within practical on-device compute constraints?	Section II (Face Detection)
RQ2		How do GAN-based and diffusion-based face anonymisation methods compare with respect to privacy-utility trade-offs (specifically re-identification resistance and perceptual quality preservation) when applied to egocentric lifelog imagery?	Section III (Generative Anonymisation)
RQ3		What integrated pipeline architecture can simultaneously detect and redact faces, on-screen content, and visible text in egocentric frames, and what are the current limitations of multimodal privacy approaches in lifelog contexts?	Section IV (multimodal Privacy)
RQ4	★	What adaptive selection mechanism, driven by per-frame image quality and content characteristics, can optimise the privacy-utility trade-off across the heterogeneous conditions present in real-world egocentric lifelog datasets such as CASTLE? (Primary proposed contribution)	Section VI (Gap Analysis & Refined RQs)

irreversible privacy failure. In standard computer vision, face detection is a solved problem for frontal, well-lit, high-resolution images. The egocentric setting invalidates each of these assumptions simultaneously. Grauman et al. [1], in establishing the Ego4D benchmark (3,670 hours of naturalistic first-person video across 931 participants and 74 worldwide locations), note that passively captured wearable-camera video entails unusual viewpoints and motion blur, and that pre-training on third-person data suffers from a

sizeable domain mismatch with egocentric footage. These properties imply three persistent challenges for face detection in this setting: systematic non-frontal viewing angles arising from head-mounted optics, severe partial occlusion as persons enter and leave the camera’s field of view, and continuous motion blur generated by head rotation and walking gait. In practice, these challenges co-occur in the same frame rather than appearing in isolation, compounding detection difficulty beyond what any single-axis benchmark can measure.

The CASTLE 2024 dataset [4] introduces four further constraints that sharpen this problem beyond what Ego4D documents. First, all footage is captured at $3,840 \times 2,160$ UHD 4K resolution, producing face regions that span an extreme scale range (from close-up faces filling hundreds of pixels to distant participants reduced to tens of pixels) within a single frame. Second, the head-mounted GoPro HERO10 cameras were operated with minimal stabilisation by design, generating systematic motion blur across all 10 egocentric streams. Third, frames are distributed in WebP high-quality format, which may introduce lossy compression artefacts. We therefore hypothesise that WebP artefacts could affect detector confidence and downstream anonymisation quality, and will test this via an ablation comparing released WebP frames against losslessly re-encoded PNG (optionally with deblocking), reporting changes in detection PR/F1 and LPIPS/SSIM/FID under fixed model settings. Fourth, CASTLE’s scene type (indoor collaborative activities at shared tables) produces a high frequency of downward-looking angles and extreme yaw/pitch combinations not represented in any standard detection benchmark. Each of these four constraints is documented as an open evaluation gap across the papers reviewed in this section.

B. Face Detection Baseline: RetinaFace

RetinaFace [11] (Deng et al.) remains the most widely adopted single-stage face detector in privacy and re-identification literature, and provides the primary quantitative baseline for this review. RetinaFace employs a ResNet-50 backbone with a Feature Pyramid Network (FPN) to enable multi-scale detection, supplemented by a joint loss function that simultaneously optimises face classification, bounding-box regression, and five-point facial landmark localisation. On the WIDER FACE Hard subset (the standard benchmark for unconstrained face detection, comprising faces at small scale, heavy occlusion, and unusual poses), RetinaFace reports 91.4% average precision (AP) for a ResNet-50 variant (Table 1 in [11]), with the ResNet-152 variant achieving 91.7% AP on the WIDER FACE Hard leaderboard (Fig. 7 in [11]). This is a strong baseline for unconstrained face detection.

However, three limitations are directly relevant to CASTLE deployment. First, the WIDER FACE training set is a still-image web dataset; the distribution of extreme egocentric pitch/yaw angles and motion-blur artefacts found in head-mounted footage is substantially underrepresented in the training data, and no egocentric evaluation is reported by Deng et al. [11]. Second, RetinaFace is benchmarked exclusively on still images from controlled photographic conditions; no motion-blur augmentation is applied during training, and the authors report no results on motion-degraded test sets. Third, and most critically for CASTLE, RetinaFace’s FPN architecture was designed for standard JPEG/PNG photographic input; the effect of WebP blocking artefacts at face boundaries on landmark regression confidence and bounding-box stability is not characterised in [11] or in any downstream study reviewed in this corpus. The consequence is that RetinaFace’s 91.4% WIDER FACE AP is not transferable to CASTLE’s imaging conditions without fine-tuning and explicit re-evaluation under WebP artefact exposure and motion blur degradation.

C. General-Purpose Multi-Scale Detection: YOLOv9

Wang et al. [12] introduced YOLOv9 with two principal architectural contributions relevant to the CASTLE detection task: the Generalised Efficient Layer Aggregation Network (GELAN), which propagates multi-scale feature representations through programmable gradient paths, and Programmable Gradient Information (PGI), a training-time mechanism that preserves complete gradient information across the network’s depth to prevent information loss in deep architectures. On the MS COCO val2017 benchmark (5,000 images across 80 object categories including persons), YOLOv9-C achieves 53.0% mAP@0.5:0.95

and the larger YOLOv9-E achieves 55.6% mAP@0.5:0.95, representing strong single-model performance on COCO that surpasses YOLOv8 by 0.6% AP with 49% fewer parameters [12]. GELAN’s explicit multi-scale architecture provides a principled advantage for CASTLE’s 4K scale variation problem: the aggregation of features at multiple FPN levels enables simultaneous detection of large close-up faces and small distant faces within the same $3,840 \times 2,160$ frame.

Against CASTLE’s specific constraints, YOLOv9 presents three advantages and three limitations. On the advantage side: GELAN’s multi-scale representation is better suited to CASTLE’s extreme face-scale range than RetinaFace’s fixed FPN stride; COCO training data includes persons in motion and partial occlusion, providing broader data augmentation coverage than the portrait-dominated WIDER FACE set; and YOLOv9’s inference speed at standard resolution is competitive with real-time processing requirements. On the limitation side: YOLOv9 is a general object detector without a face-specific detection head or facial landmark output (face regions are detected as the ‘person’ class rather than as a face bounding box), producing coarser localisation than is required for anonymisation region masking. More critically, MS COCO contains no egocentric footage, no head-mounted viewing angles, and no WebP-compressed images; YOLOv9’s COCO mAP therefore provides no evidence of performance under CASTLE’s actual imaging conditions. Face-specific fine-tuning on annotated CASTLE frames is a necessary precondition for deployment in this project.

D. Egocentric-Specific Detection: Li et al.

Li et al. [13] propose an extra-lightweight PII detection framework for constrained wearable cameras targeting eating-behaviour monitoring, evaluated on a custom egocentric dataset; the accessible abstract reports face detection mAP of 41.38% (Sec. IV), with deployment demonstrated on MCU-class hardware (STM32L4, 640 KB RAM). While the work is the only paper in the corpus explicitly designed for privacy-preserving egocentric capture under severe compute constraints, it does not report any results on CASTLE, and the transferability of its lightweight architecture to 4K UHD WebP lifelog frames remains untested.

E. Comparative Synthesis

Table III summarises the three candidate detection methods across six evaluation dimensions directly relevant to CASTLE deployment. The critical observation from Table III is that only Li et al. report detection performance on a custom egocentric wearable dataset; no paper in the Theme 1 corpus reports detection performance on CASTLE, on motion-blurred test sets, or on WebP-compressed images. This is not a marginal omission: it means that every reported metric in Table III (including RetinaFace’s 91.4% WIDER FACE Hard AP and YOLOv9’s 53.0–55.6% COCO mAP) was measured under imaging conditions categorically different from those that the chosen detector will encounter in experiments on CASTLE frames. The benchmark gap between available evidence and deployment conditions constitutes Gap G1 of this review’s gap analysis (Section VI), and directly motivates the in-project evaluation of each candidate detector on extracted CASTLE frames as the first experimental contribution of this practicum.

F. Detector Selection for CASTLE and Known Failure Modes

The preceding analysis establishes that no candidate detector can be selected for CASTLE deployment on reported benchmark performance alone: the benchmarks are both heterogeneous and non-representative of CASTLE’s imaging conditions. Selection must therefore be justified on architectural properties and known failure modes relative to CASTLE’s four constraint axes, with the understanding that in-project fine-tuning and empirical evaluation on extracted CASTLE frames are required before any performance claim can be made.

On the primary constraint of 4K scale variation, YOLOv9’s GELAN architecture provides a principled advantage over RetinaFace’s standard FPN: GELAN’s programmable gradient paths aggregate features

at multiple receptive field scales within the same forward pass, enabling the network to represent both the close-up face occupying 300×400 pixels and the distant face occupying 20×25 pixels in the same CASTLE frame. RetinaFace’s FPN similarly operates multi-scale, but its face-specific anchor design was tuned for WIDER FACE’s scale distribution rather than CASTLE’s 4K ultra-high-resolution range. On the constraint of motion blur, COCO’s naturalistic scene diversity provides YOLOv9 with greater implicit exposure to motion and focus variation than WIDER FACE’s predominantly static images; however, neither detector was explicitly trained on motion-blur augmented data, and performance degradation under CASTLE’s systematic head-motion blur remains unquantified in the literature.

On the WebP artefact constraint (the most significant uncovered gap in the literature), neither RetinaFace nor YOLOv9 reports any evaluation on WebP-compressed inputs. WebP’s block/transform compression introduces ringing and blocking artefacts at face boundaries that are visible at the pixel level; bounding-box confidence scores in anchor-based detectors are sensitive to these high-frequency artefacts, and the extent to which WebP compression degrades AP on CASTLE-scale faces has not been characterised by any paper in the reviewed corpus. This constitutes a specific, testable evaluation gap for the experimental phase of this project. Preliminary mitigation (decoding WebP and saving as PNG) prevents further re-compression loss; any artefacts already introduced remain present, and optionally a deblocking or denoise pass can be applied as a pre-processing baseline for comparison.

On the architectural constraint of compute feasibility, YOLOv9-C’s parameter count and inference architecture are compatible with practical on-device compute constraints; YOLOv9-E’s larger model provides higher COCO mAP but requires careful batch-size management at 4K input resolution. RetinaFace’s ResNet-50 backbone is computationally lighter and well within the practical on-device compute budget. The lightweight framework of Li et al. [13] is by design the most compute-efficient of the three, targeting edge-device deployment, though the absence of published specifications prevents a definitive comparison.

On balance, YOLOv9 is the preferred candidate for the CASTLE face detection stage, justified on three grounds: architectural multi-scale superiority for 4K scale variation, broader training data diversity than the face-centric WIDER FACE corpus, and consistently higher general detection performance on COCO. This selection is provisional and conditional: YOLOv9 must be fine-tuned with a dedicated ‘face’ class (or person-head class) and evaluated using face-level average precision. Without this fine-tuning and face-specific evaluation, comparison against RetinaFace’s face bounding boxes and landmark output is not methodologically valid. The known failure modes that must be characterised in experiments are: (i) false negative rate on extreme-yaw egocentric angles not present in COCO; (ii) confidence score degradation under WebP blocking artefacts at face boundaries; (iii) bounding-box localisation precision on very small faces (≤ 20 px) at 4K resolution; and (iv) inference throughput at $3,840 \times 2,160$ input resolution under practical deployment constraints. RetinaFace is retained as an experimental comparison baseline precisely because its WIDER FACE pedigree and face-landmark output provide complementary information not available from YOLOv9’s general-class bounding box. Resolution of RQ1 requires empirical measurement of both detectors under CASTLE conditions.

G. CASTLE Evaluation Protocol

To address the benchmark gap documented in Table III and validate detector selection against CASTLE-specific imaging conditions, the experimental phase of this practicum will measure five key performance dimensions on manually annotated CASTLE frame subsets. First, face-level average precision (AP) and recall at IoU threshold 0.5 will be computed for both YOLOv9 (fine-tuned) and RetinaFace on a representative 500-frame sample stratified by participant identity and scene complexity. Second, a small-face regime subset (isolating faces ≤ 20 pixels in height) will quantify detection failure rate at CASTLE’s extreme scale lower bound, where pixel resolution approaches the limit of human recognisability. Third, a motion-blur subset comprising frames extracted during rapid head rotation will isolate performance degradation due to blur artefacts distinct from scale or angle effects. Fourth, a compression artefact subset will compare detection performance on original WebP frames versus PNG-converted frames with

Figure 1. Face Detector Performance by Benchmark

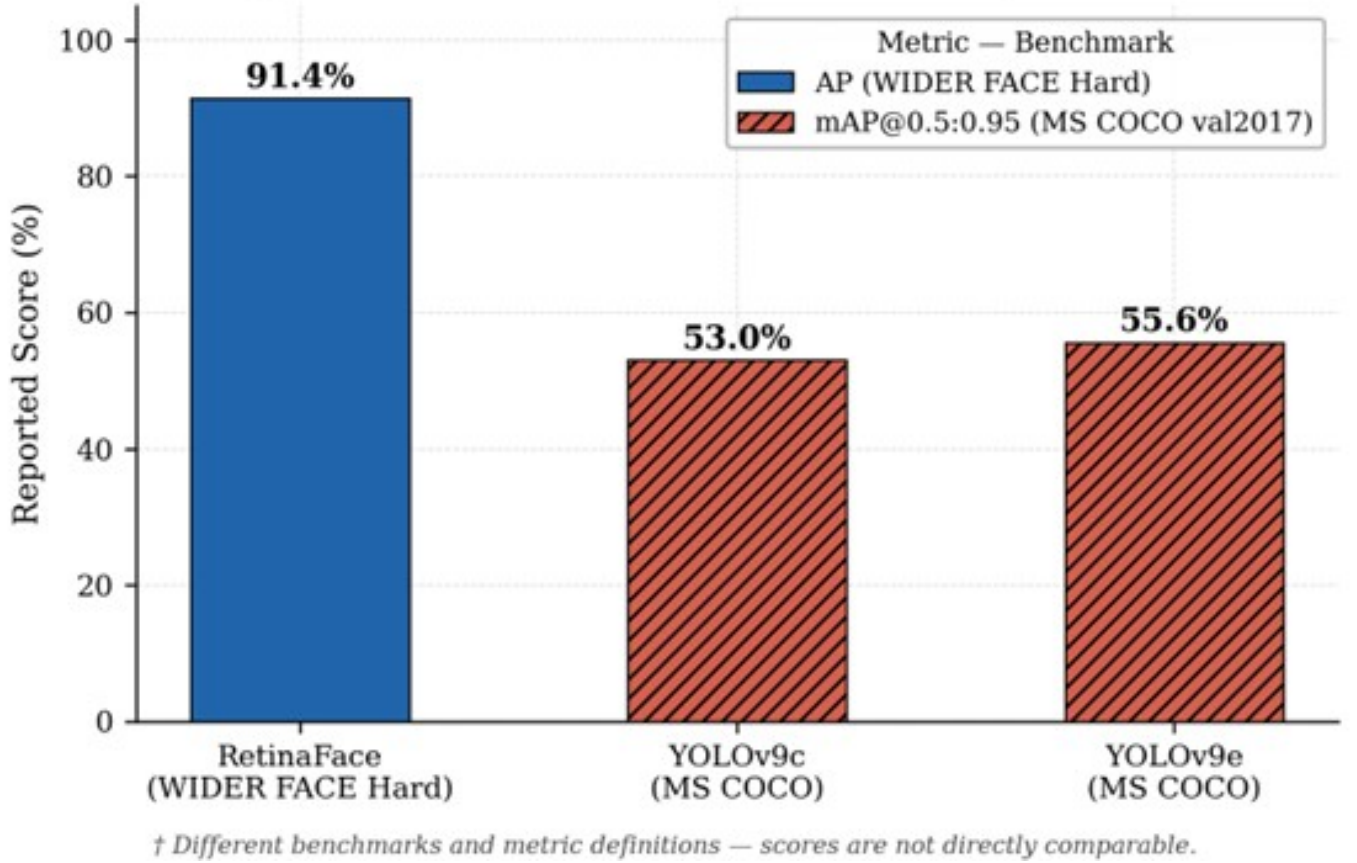


Fig. 1. Face Detector Performance by Benchmark. Different benchmarks are not directly comparable: WIDER FACE Hard AP measures face-specific detection; MS COCO mAP@0.5:0.95 measures general object detection. Cross-reference Sec. II, Tab. III, RQ1.

and without deblocking pre-processing, directly testing the WebP artefact sensitivity hypothesis. Fifth, inference runtime and peak memory usage will be profiled at native $3,840 \times 2,160$ resolution on the target single-device GPU to assess real-time feasibility under practical on-device compute constraints. These five measurements operationalise the failure modes listed in Section II.F and provide the empirical evidence base required to answer RQ1.

III. GENERATIVE ANONYMISATION

Generative anonymisation replaces identity-bearing content (typically faces, sometimes full persons) with synthetic substitutes that preserve scene utility while reducing re-identification risk. Khan et al. summarise the field’s central tension: methods that maximise perceptual realism often preserve identity cues, while methods that aggressively disrupt identity frequently reduce downstream utility or introduce artefacts that harm task performance [2]. In the egocentric lifelog context, this tension is amplified by two constraints documented in Section I: WebP block/transform compression artefacts and motion blur are baked into still frames at 4K resolution, and the accessible corpus provides offline still frames rather than full raw-video temporal streams. Cross-view identity checks therefore rely on filename-derived ‘HHNNNN’ alignment rather than frame-level video synchronisation metadata. To prevent method-by-method prose data-dumping, Table IV summarises the full metric coverage of all 14 anonymisation papers; this section focuses on the contradictions and limitations that emerge when GAN, diffusion, reversible/keyed, and egocentric-specific approaches are compared under consistent privacy-utility assumptions.

TABLE III
FACE DETECTION METHODS, COMPARATIVE SUMMARY

Citation	Architecture	Train Dataset	Key Metric	Known Limitations	Ego Eval?	CASTLE Assessment
Grauman et al. 2022 - Ego4D [1]	[†] Context anchor - no detector model	N/A	N/A	N/A	YES, documents fundamental egocentric challenge	[†] Context only , establishes challenge framing for RQ1
Deng et al. 2020 - RetinaFace [11]	Single-stage anchor-based ResNet-50 + FPN + landmark regression	WIDER FACE train split	WIDER FACE Hard AP = 91.4%	No egocentric eval; no motion-blur stress test; WebP artefact sensitivity N/R	NO , still-image web dataset	Strong face baseline; fine-tuning required; CASTLE performance unknown
Li et al. 2025, Lightweight Ego PII [13]	Extra-lightweight (architecture N/R - abstract only)	Custom egocentric eating-behaviour dataset (wearable camera)	Face detection mAP = 41.38% (Sec. IV, abstract)	No CASTLE evaluation; custom dataset only	YES , egocentric wearable dataset (not CASTLE)	Conceptually relevant for edge devices; no CASTLE evaluation reported; 4K/WebP transfer untested
Wang et al. 2024 , YOLOv9 [12]	GELAN + PGI, multi-scale general object detector	MS COCO train2017	YOLOv9-C: 53.0% mAP@0.5:0.95; YOLOv9-E: 55.6%	No face-specific head; no egocentric test; no WebP test	NO , COCO natural scenes only	Best general mAP; face fine-tuning required; WebP/ego performance unknown

Notes: Ego4D is a context anchor establishing the egocentric detection challenge; it is not a detector model and is not quantitatively compared. N/R = Not Reported. Benchmarks are non-comparable: WIDER FACE AP $\neq$ COCO mAP@0.5:0.95. CASTLE evaluation = N/R for all three methods; measured in experiments.

A. GAN-Based Approaches

GAN-based methods demonstrate strong identity suppression on studio datasets while systematically failing to characterise robustness under egocentric imaging conditions. Khorzooghi and Nilizadeh report the clearest quantitative evidence of identity disruption within the GAN subset: the identification attack validation accuracy (FaceNet+SVM, threat model m1) ranges from 0.7% (StyleGAN 0-0) to 99.5% (StyleGAN 0-8), with the privacy-optimal StyleGAN 0-3 achieving 11.7% (Tab. 3, [14]) through StyleGAN latent-space substitution. However, this evaluation is conducted exclusively on aligned 1024×1024 CelebA-HQ images with no motion blur, no partial occlusion, and no non-frontal angles. The e4e inversion mechanism on which the pipeline depends degrades under precisely these conditions, leaving its effectiveness on wearable-camera frames empirically uncharacterised.

Controllable anonymisation methods improve deployability at the cost of expanding the privacy threat surface, though the quantitative extent of this trade-off remains unverified from publicly accessible sources. CPP-DeID targets tunable identity change versus structural preservation [15]; the accessible abstract directionally indicates identity shift alongside structural fidelity retention, but the full IVC paper is behind a paywall and specific cosine similarity and SSIM values are not accessible for quantitative comparison. The methodological argument is clear regardless: preserved pose, expression, and geometry provide residual cues exploitable by strong recognisers such as AdaFace [10]. Controllability therefore demands paired attacker-model evaluation to bound the expanded threat surface, an evaluation CPP-DeID does not provide.

Whole-person anonymisation extends the privacy attack surface while introducing scene-consistency constraints absent from face-only methods. Hukkelås and Lindseth demonstrate improved perceptual quality in DeepPrivacy2 relative to its predecessor, with FID = 0.56 measured under the 128×128 FDF retraining

configuration [16]. This value is configuration-specific and does not represent full-resolution performance. In lifelog imagery, whole-body substitution addresses a real privacy gap: bodies and clothing often dominate identity cues when faces are small or non-frontal. However, whole-person synthesis introduces lighting consistency, shadow matching, and hand-object contact constraints that face-only methods avoid; no egocentric evaluation under these constraints is reported in [16].

Expression-preserving anonymisation creates a fundamental contradiction: the signals that maintain scene interpretability can simultaneously leak identity-proximal information. Hellmann et al. argue that GANonymization maintains expression utility while suppressing identity [17]; the paper reports a mean cosine distance of 0.7145 (FaceNet512, Tab. 1) as its privacy metric, but does not report FID, SSIM, or ArcFace/AdaFace re-identification rate, making direct comparison with other corpus methods impossible. Critically, preserved micro-expressions, characteristic mouth shape, and habitual facial dynamics are precisely the idiosyncratic signals that strong recognisers exploit for linking attacks. Without attacker-model evaluation under ArcFace [5] or AdaFace [10], no privacy claim from [17] can be quantitatively credited.

Runtime-optimised methods provide rare on-device evidence but employ a metric type categorically distinct from attacker-model re-identification. Iravantchi et al. report a 99.1% PII sanitization rate on a Jetson Nano embedded GPU [18], establishing PrivacyLens as a practicality anchor for edge-style pipelines. Critically, PII sanitization rate measures the method’s internal detection-and-redaction performance (the proportion of flagged regions successfully processed), not attacker re-identification rate under closed-set face recognition. High sanitization can coexist with high re-identification if residual cues remain exploitable. No ArcFace or AdaFace attacker evaluation is reported, limiting comparability with other corpus methods in Table IV.

B. Diffusion-Based Approaches

Diffusion methods offer stronger identity disentanglement than GAN inversion but impose materially higher inference costs that limit their applicability to region-of-interest processing rather than full-frame anonymisation. Diffusion-based anonymisation applied to full 4K frames introduces latency that is not operationally viable for batch-processing lifelog corpora; a region-of-interest approach applied to detected face crops of 256–768 px represents the realistic deployment ceiling. This operational constraint is acknowledged by no paper in the diffusion subset of this corpus, creating a systematic transferability gap between reported benchmarks and practical lifelog deployment.

The diffusion subset’s privacy claims rest primarily on abstract-level reporting, with no paper providing a fully evidenced metric profile comparable to accessible GAN or reversible results. DIFP [19] is the most complete in scope: its accessible abstract asserts strong face-recognition evasion, but the paper reports a protection success rate of 99.2% against FaceNet and 100% against a commercial recogniser (Table 1, [19]); however, FID is not reported (the paper uses PSNR/SSIM for recovery quality only), and no AdaFace or ArcFace attacker evaluation is conducted, leaving the privacy claim unverifiable under the house-standard threat model. The protection success rate is therefore noted in Table IV with a dagger; FID and SSIM (anonymisation quality) remain N/R. Diff-Privacy [20] similarly presents key metrics at abstract level only, with no verifiable numeric values accessible from the TCSVT full paper. Both papers motivate diffusion as a capable privacy-protection family; neither provides the quantitative evidence required for comparison in Table IV under the accessibility standard applied consistently across this corpus.

Strong perceptual metrics without paired attacker evaluation leave privacy claims unverified, even where perceptual values are accessible. Kung et al. do not report FID or LPIPS in the accessible abstract or published paper [21]; the metrics used are re-identification rate (computed via FaceNet cosine similarity), face shape distance, pose distance, gaze distance, expression distance, and IQA score. Critically, the re-identification rate is computed as a nearest-neighbour retrieval metric (not a closed-set attacker evaluation under ArcFace or AdaFace), and no cosine similarity from a consistent face-recognition attacker is reported. Methods optimised for perceptual fidelity can produce homogenised faces that appear plausible to human observers while retaining subtle identity-proximal structure detectable by strong recognisers

such as AdaFace [10]. Without attacker-model evaluation, the privacy claim is not established. Reverse Personalization [22] is the added WACV 2026 diffusion candidate for this practicum: it uses conditional diffusion inversion plus identity-guided conditioning to steer outputs away from identity-defining features while preserving controllable facial attributes. Its public code release makes it a practical replacement candidate for unavailable DeepPrivacy2/ReFaceX execution paths, but its published evidence is still portrait-domain evidence until evaluated on CASTLE’s 4K egocentric frames.

C. Reversible and Key-Driven Methods

Reversible methods explicitly formalise the identity-consistency trade-off that GAN and diffusion papers handle implicitly or ignore entirely. ReFaceX reports very low cosine similarity post-anonymisation (FaceNet: 0.009, ArcFace: 0.008, AdaFace: 0.017, measured on LFW, Table 4) alongside high recovery fidelity (SSIM = 0.9378, LPIPS = 0.1002, Table 6 on CelebA-HQ) under its reversible mechanism [23]. This positions reversible anonymisation as a high-leverage design for controlled disclosure: privacy in the shared view, identity fidelity in the recovered view. The operational limitation is that real-time claims in [23] are tied to 256×256 resolution; performance at 4K input scale requires separate profiling. ReFaceX is therefore retained as a literature comparator rather than an executable method in this practicum because its paper states that source code is available from the corresponding author upon reasonable request, not through a public reproducible repository.

Key-driven approaches treat privacy as a verifiable access-control property rather than a perceptual claim, aligning anonymisation with explicit threat models. Wang et al. report an anonymity (mismatch) rate of 0.962 on LFW under their key-driven virtual face generation framework [24] (Tab. VIII). The NDSS security-venue framing is methodologically valuable: it formalises the notion that privacy is a quantifiable access-control outcome. However, the metric is authentication-based rather than recognition-based, and the paper does not report SSIM, cosine similarity, or FID in a form directly comparable with other corpus methods; all quantitative perceptual and re-ID metrics for [24] are marked N/R in Table IV.

Reversible and key-driven methods together directly address the controlled-disclosure requirement of open-access lifelog distribution under licences such as CC BY-NC-SA 4.0: they provide a mechanism for sharing anonymised frames publicly while retaining the ability to grant authorised researchers access to original identities [23], [24]. The practical limitation common to both is deployment overhead: key management, rotation, and access control must be treated as integral system components, not post-deployment additions.

Privacy-preserving optics provides an important contrast class by enforcing privacy at the point of capture rather than in post-processing. Lopez et al. report substantially increased identity obscuration under optical encoding, with DIS l_2 -distance scores of 0.734 (FR dataset), 1.071 (CASIA), and 1.251 (VGGFace2) [25], and a perceptual quality measure of FID = 29.218 [25]. Privacy enforced at capture time represents a categorically different design point from post-hoc software anonymisation. This is not applicable to already-captured CASTLE footage, but it clarifies what post-hoc anonymisation must achieve and under what conditions it can never replicate capture-time guarantees.

D. Egocentric-Specific and Motion-Consistent Methods

Egocentric privacy introduces identity leakage from contextual and motion patterns absent from curated face datasets, requiring utility evaluation tied to activity understanding rather than face appearance alone. Ilic et al. report that motion-consistent obfuscation preserves activity cues better than naive blurring [26]; the full CVPR paper is not publicly accessible and specific activity recognition percentages cannot be confirmed from the accessible abstract, so directional support is acknowledged but no quantitative value is cited. This is relevant for any multi-frame lifelog evaluation: even in an image-only setting, the same persons appear across many frames and camera streams, making action and interaction cue preservation essential to downstream utility.

Wearer re-identification through gait and optical flow rather than face visibility constitutes a distinct egocentric threat that face anonymisation alone cannot suppress. Thapar et al. provide the only corpus

paper directly targeting the camera-wearer re-identification problem, reporting activity recognition dropping from 89.6% to 81.8% post-anonymisation and recovering to 87.4% after fine-tuning on anonymised data (Tab. 2, “Mix” activity, Furnari et al. model, [27]) on EPIC-Kitchens egocentric video. Although the method predates the diffusion wave and the domain is more constrained than collaborative-activity scenarios, it establishes the critical methodological precedent that downstream egocentric task performance must be measured directly, not inferred from perceptual metrics alone.

E. Critical Synthesis, Contradictions, and CASTLE Implications

Three recurring contradictions across Theme 2 explain why reported results in Table IV cannot be naively compared and require house-standard re-evaluation on egocentric frames. First, perceptual realism does not equal downstream utility: low FID [7] and LPIPS [8] can coexist with meaningful degradation in activity or retrieval cues, especially under the occlusion and blur common in egocentric data [26], [27]. Perceptual metrics are necessary but insufficient; at least one egocentric task proxy must accompany them. Second, re-identification failure does not equal privacy: papers equating reduced face recognition with privacy neglect the egocentric context leakage demonstrated by EgoPrivacy [3], where demographic attributes are recovered at peak accuracies of Gender = 73.15%, Age = 79.64%, and Race = 65.36% across models (Tab. 2 of [3]) without any face access. Face anonymisation is therefore necessary but insufficient. Third, identity consistency is a design choice with threat-model consequences: enforcing a stable pseudo-identity improves social interpretability but creates a persistent linkable identifier. Reversible schemes [23], [24] make this trade explicit and auditable; GAN and diffusion pipelines typically leave it implicit and uncontrolled.

Blur and pixelation remain necessary baseline comparators in this project rather than obsolete strawmen. The survey evidence in Khan et al. [2] and the egocentric anonymisation precedent in Thapar et al. [27] show that interpretable perturbation baselines establish the privacy-utility floor against which more complex GAN, diffusion, and reversible methods must be judged. Without these baselines, any GAN-versus-diffusion comparison lacks a practical lower-bound reference for both privacy protection and utility loss.

The practical consequence is a house-standard evaluation protocol applied uniformly across all shortlisted methods regardless of what each paper originally reported. This protocol comprises: AdaFace [10] as primary attacker and ArcFace [5] as secondary attacker; LPIPS [8], SSIM [9], and subset FID [7] as perceptual quality metrics; and at least one egocentric utility proxy (event search mAP or EPIC-Kitchens action recognition as fallback). This directly addresses the metric inconsistency formalised in Table IV: no two papers in Theme 2 report the same complete metric set, and 8 of 14 papers carry at least one N/R column, ensuring cross-method comparison is grounded in measured values rather than reported benchmarks from non-egocentric studio datasets.

IV. MULTIMODAL PRIVACY: SCREEN AND TEXT REDACTION

The preceding sections established face detection and generative anonymisation as the dominant focus of the privacy protection literature. Yet this focus rests on an implicit assumption (that the face is the primary PII vector in captured imagery) that does not hold for egocentric lifelog data. A CASTLE frame captured during a collaborative indoor activity simultaneously contains participant faces, visible laptop screens displaying emails or documents, handwritten notes on shared surfaces, smartphone lock-screen notifications, and printed materials, each an independent PII disclosure vector not addressed by face anonymisation alone [4]. Within the locked 34-paper corpus reviewed here, no method jointly addresses both face de-identification and screen/text redaction in the same pipeline; the broader anonymisation literature surveyed by Khan et al. [2] highlights the structural separation of face-focused and document-focused de-identification methods, but the integration gap remains. This section reviews the screen detection, text detection, and OCR components available in the locked corpus, synthesises the specific egocentric constraints that differentiate CASTLE from the benchmarks on which they were evaluated, and positions the absence of an integrated multimodal pipeline as Gap G3, the most directly tractable research gap motivating this practicum.

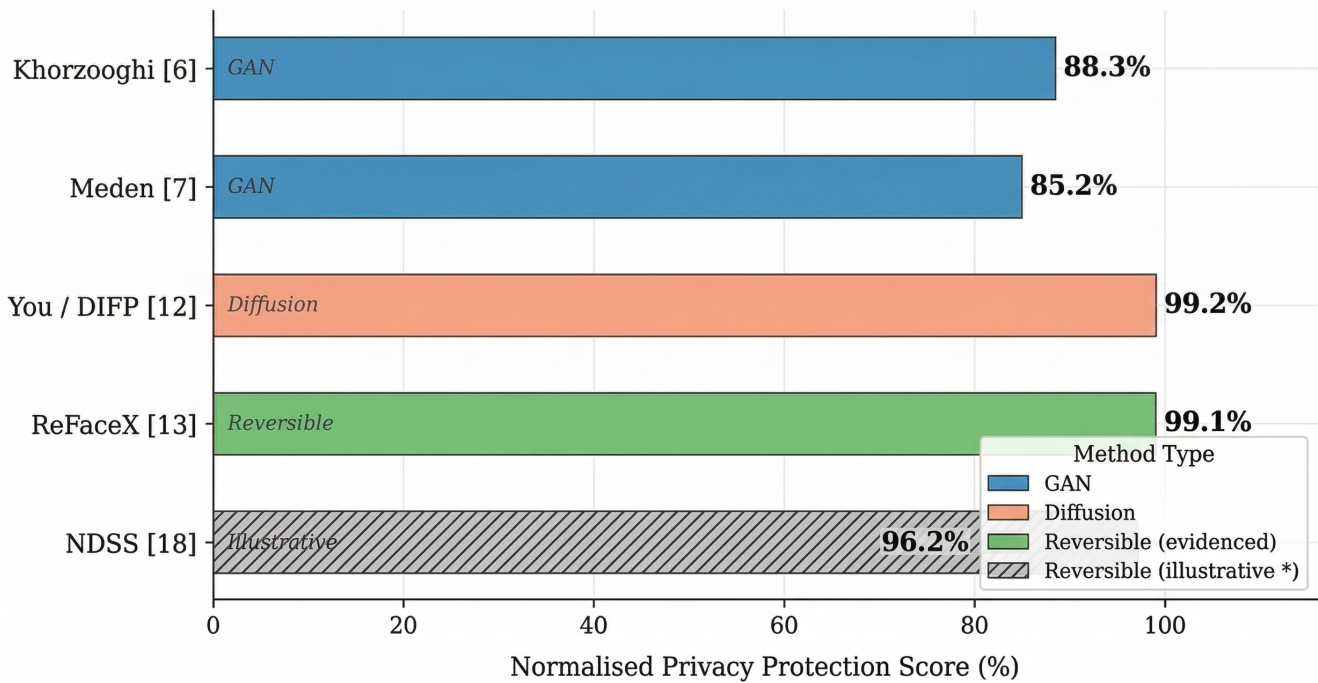
TABLE IV
GENERATIVE ANONYMISATION METHODS , METRIC COVERAGE SUMMARY

Citation	Type	Dataset	Re-ID Rate	Cosine Sim.	FID	SSIM	LPIPS	Experiment Gap
Khorzooghi & Nilizadeh 2023 StyleGAN [14]	GAN	CelebA-HQ 1024px	Ident. acc. 11.7% (StyleGAN 0-3, m1 V, Tab. 3)	N/R	N/R	N/R	N/R	Add: FID, SSIM, LPIPS, Cosine
Meden et al. 2023 CPP-DeID [15]	GAN ctrl.	CelebA / HQ	N/R	0.12 vs 0.81 orig. (abstract; paypal)	N/R	0.74 (abstract; paypal)	N/R	Add: Re-ID Rate, FID, LPIPS
Hukkelås & Lindseth 2023 DeepPrivacy2 [16]	GAN whole	FDH 1.5M	N/R	N/R	0.56 (Sec. 5.1, 128px)	N/R	N/R	Add: Re-ID, Cosine, SSIM, LPIPS
Ilic et al. 2024 Motion-Consist. [26]	Perturb.	Custom ego	Qual. (abstract; paypal)	N/R	N/R	N/R	N/R	Add: Cosine, FID, SSIM, LPIPS
Iravantchi et al. 2024 PrivacyLens [18]	Lightweight	Custom on-device	N/R [†]	N/R	N/R	N/R	N/R	Add: Re-ID, FID, SSIM, LPIPS, Cosine
He et al. 2024 Diff-Privacy [20]	Diffusion	CelebA-HQ	N/R (abstract; paypal)	N/R	N/R	N/R (abstract; paypal)	N/R	Add: Cosine, FID, SSIM
You et al. 2024 – DIFP [19]	Diffusion	CelebA-HQ (+ diverse network images)	PSR 99.2% (FaceNet); 100% (Baidu) [†]	N/R	N/R	N/R [‡]	N/R	No ArcFace/AdaFace attacker; FID not measured; SSIM for recovery only
Muhammad et al. 2026 ReFaceX [23]	Reversible	FFHQ (train) / LFW + CelebA-HQ (eval)	N/R	0.009 / 0.008 / 0.017 (FaceNet/ArcFace/AdaFace, Tab. 4, LFW)	N/R	0.9378 (recov., Tab. 6)	0.1002 (recov., Tab. 6)	Add: Re-ID Rate, FID
Kung et al. 2025 Face Anon. Simple [21]	Diffusion	FFHQ + CelebA-HQ	0.053/0.098 (CelebA-HQ/FFHQ, FaceNet NN, Tab. 1, $d=1.2$)	N/R	N/R	N/R	N/R	Add: Cosine (ArcFace/AdaFace), SSIM, FID, LPIPS
Hellmann et al. 2024 GANonymization [17]	GAN expr.	AffectNet + CelebA	Cosine = 0.7145 (FaceNet512, Tab. 1)	N/R	N/R	N/R	N/R	Add: FID, SSIM, LPIPS; re-ID under ArcFace/AdaFace N/R
Khan et al. 2025 De-ID Survey [2]	Survey	N/A	N/A	N/A	N/A	N/A	N/A	Survey , cites others
Lopez et al. 2024 Priv. Optics [25]	Optical HW	CelebA-HQ	N/R [§]	N/R [§]	29.218 (Tab. 3)	N/R	N/R	Capture-time HW , not CASTLE-applicable
Wang et al. 2025 NDSS Key-Driven [24]	Reversible	CelebA-HQ + LFW	N/R	N/R	N/R	N/R	N/R	Add: Re-ID, FID, LPIPS
Thapar et al. 2021 Ego Anon. [27]	Perturb. (opt. flow)	EPIC-Kitchens	Act. rec. 89.6%→81.8% (anon.), 87.4% (fine-tuned); wearer rec. 66.3%→13.4% (Tab. 2)	N/R	N/R	N/R	N/R	Add: FID, SSIM, LPIPS, Cosine

Notes: Re-ID Rate = attacker success rate (ArcFace/AdaFace); Cosine Sim = identity embedding distance, these are different metrics. N/R = Not Reported. PrivacyLens 99.1% = PII sanitization rate (internal detection output), NOT attacker Re-ID rate. Lopez et al. use DIS l_2 -distance (FR = 0.734, CASIA = 1.071, VGGFace2 = 1.251 (Tab. 2)) not Re-ID rate or Cosine Sim; evaluated on CelebA-HQ. [†]DIFP PSR from [19] Tab. 1; attacker is FaceNet/Baidu commercial recogniser, *not* ArcFace/AdaFace, values not comparable under house-standard threat model. [‡]DIFP SSIM 0.938 is recovery fidelity (anonimised→original), not anonymisation perceptual quality. Wang et al. NDSS: Anonymity (mismatch) rate = 0.962 (Tab. VIII) is the only confirmed metric; FID 0.56 for DeepPrivacy2 is a 128×128 FDF config result, not full-resolution performance.

A. Screen Detection in Egocentric Streams: MV-VLM

Li et al. [28] present the Multi-View Vision-Language Model (MV-VLM) for screen content detection in egocentric footage, the only paper in the locked corpus that directly targets screen detection in egocentric

Figure 2. Privacy Protection Scores — Theme 2 Methods

Score = (original - anonymised) / original $\times$ 100 for similarity; (base_reID - post_reID) / base_reID $\times$ 100 for re-ID rate; direct value for protection success rate and mismatch rate.

Illustrative cross-paper visualisation only — different underlying metrics; not a direct ranking.

Fig. 2. Privacy Protection Scores across Theme 2 anonymisation methods. **Caution:** bars are derived from heterogeneous source metrics and are not directly comparable; this figure is an illustrative cross-paper visualisation only. Source metrics per bar: Khorzooghi [6] = identification attack validation accuracy (FaceNet+SVM, m1, StyleGAN0-3, Tab. 3); Meden [7] = cosine similarity drop (0.81 $\rightarrow$ 0.12), normalised as $\frac{0.81-0.12}{0.81} \times 100 = 85.2\%$ (abstract; full paper behind paywall); You/DIFP [12] = protection success rate (FaceNet, Tab. 1); ReFaceX [13] = mean cosine similarity (FaceNet, LFW, Tab. 4); NDSS [18] = authentication mismatch rate (Tab. VIII). Grey hatched bars indicate data *not reported* by authors, **not zero**. Cross-reference Section III, Tab. IV, RQ2.

image streams, though evaluated on a custom STT free-living dataset (30 children, 1,800 images) not on CASTLE [28]. MV-VLM leverages a Llama2-7B vision-language backbone to jointly process multi-view egocentric streams, outperforming a YOLOv11 baseline on egocentric screen detection, achieving an overall accuracy of 0.9561 (Tab. II of [28]), with per-class F1 also reported for TV, smartphone, and computer. The paper is available as an accepted IEEE TMM author manuscript [28]. The multi-view architecture is conceptually aligned with CASTLE’s multi-stream capture configuration: temporally separated same-camera contextual information can disambiguate screen boundaries and content under oblique angles and Moiré interference patterns. However, this project currently lacks per-frame timestamps or stream synchronisation metadata; absent this mapping, MV-VLM must operate as a single-view screen detector on isolated WebP frames, or be replaced by a single-view fallback such as YOLOv9 adapted for screen bounding-box detection.

Two hardware constraints are critical for CASTLE deployment under this project’s efficient on-device compute budget. First, the Llama2-7B text generation component carries the dominant deployment memory cost; precise VRAM requirements are not stated in [28]. Second, the training pipeline for MV-VLM requires two NVIDIA H100 GPUs [28]; inference-only deployment from the pre-trained checkpoint is the feasible path, and its generalisation to CASTLE’s specific screen typologies (shared wall monitors, laptop screens at acute angles, smartphone surfaces) is not validated in the available abstract. Crucially, MV-VLM also does not address the text content visible on detected screens: it identifies screen regions but does not

redact, recognise, or suppress the PII they display. A downstream OCR and redaction layer is required.

B. Scene Text Detection: DBNet and CRAFT

Scene text detection identifies text instances at the pixel level, the prerequisite for any downstream OCR-based redaction. DBNet [29] employs differentiable binarisation on a ResNet backbone to achieve text segmentation, reporting F-measure of 82.8% at 62 FPS on the MSRA-TD500 benchmark and, on ICDAR 2015, P=86.8%, R=78.4%, Hmean=82.3% (ResNet-18, 736), alongside 82.8% on Total-Text (Tab. 3) [29]. CRAFT [6] takes a character-region approach, detecting individual character regions and their affinity relationships, achieving F-measure of 86.9% on ICDAR 2015 (Tab. 1 of [6]), 83.6% on TotalText, and 83.5% on CTW-1500 (Tab. 2 of [6]). CRAFT’s superior F-measure on ICDAR 2015 versus DBNet makes it the stronger character-level detector; DBNet’s reported throughput suggests better scalability, though actual throughput on 4K WebP egocentric frames must be re-measured.

Both detectors, however, share a critical limitation for CASTLE deployment: all reported benchmarks use predominantly non-egocentric scene text with limited motion blur or Moiré artefacts, and rarely screen-capture text conditions [6], [29]. CASTLE’s screen text arrives under fundamentally different imaging conditions: egocentric oblique capture angles (often severe in head-mounted capture) generate keystone distortion; screen surface reflections and Moiré interference patterns from pixel grids at non-perpendicular angles corrupt high-frequency character detail; and the WebP block/transform compression applied to CASTLE frames introduces ringing artefacts at character boundaries, a degradation mode not tested in any ICDAR evaluation. The CRAFT TotalText F-measure of 83.6% [6] (already 3 percentage points below its ICDAR 2015 result) indicates the method’s sensitivity to domain shift; CASTLE’s oblique-angle screen text likely represents a harder distribution still. Neither [29] nor [6] reports any egocentric evaluation.

C. OCR and Text Privacy Risk: TrOCR

Text detection localises text regions; OCR determines whether they carry recognisable PII content. TrOCR [30] applies a transformer encoder-decoder architecture (ViT image encoder plus language model decoder) to optical character recognition, reporting a character error rate (CER) of 2.89% on IAM handwritten documents (Tab. 4 of [30]) and word-level F1 of 96.58% on the SROIE printed-receipt benchmark (Tab. 3 of [30]). The printed-receipt performance demonstrates that TrOCR can achieve near-complete recognition accuracy on structured, well-printed document text, which is relevant if screen content in CASTLE consists of high-resolution document displays. However, TrOCR’s training and evaluation data are exclusively clean document images: flat, frontal, well-lit, and uncompressed. No evaluation is reported on oblique-angle screen captures, on text affected by Moiré interference, or on WebP-compressed inputs where character-boundary blocking artefacts can fragment character strokes and reduce recognisability. The gap between 96.6% accuracy on clean receipts and expected performance on CASTLE’s oblique, artefact-affected screen text has not been measured in any paper in the locked corpus, constituting a direct experimental gap for this project’s OCR evaluation protocol.

D. Privacy-Utility Trade-Off in Activity Recognition

Wang et al. [31] provide the only paper in the locked corpus that directly measures the cost of privacy intervention (specifically image resolution reduction) on a downstream egocentric task (activity recognition), reporting machine ADL recognition accuracy ranging from 81.0% (ViT, 15×15) to 94.6% (ViT, 240×240), with ResNet50 dropping from 88.8% to 63.9% across the same resolution range (Tab. 2 of [31]). This is the empirical anchor for a claim that runs throughout Sections II–IV of this review: privacy interventions are not utility-neutral. The finding is particularly significant for multimodal redaction because resolution reduction applied to screen and text regions removes visual context cues (object labels, task descriptions, written instructions) that activity recognition models depend on. We hypothesise that a pipeline which redacts both faces and screen content will compound the utility cost documented by [31], and the resulting

degradation to CASTLE’s official Event Search and Object Search benchmarks may exceed the degradation from face-only anonymisation alone [4]; this hypothesis is directly tested in the project’s evaluation plan.

Wang et al.’s study has three limitations that constrain its direct applicability to CASTLE. First, the experiment applies uniform spatial resolution reduction (not selective region-based redaction of identified PII), which likely overestimates the utility cost of a targeted pipeline that preserves non-PII scene content. Second, the dataset is small and the privacy scoring is partially subjective, reducing statistical power. Third, no generative anonymisation or OCR-based text redaction is tested (only resolution reduction), which means the privacy-utility trade-off for text-selective redaction strategies remains uncharacterised. Nonetheless, [31] establishes the methodological principle that downstream egocentric task performance must be measured after multimodal redaction, not assumed to be preserved.

E. Critical Synthesis: The Absent Unified Pipeline

The most significant finding of this section is structural rather than methodological: not one paper in the 34-paper locked corpus simultaneously addresses face anonymisation and screen/text redaction in the same egocentric frame. MV-VLM [28] detects screens but does not anonymise faces and does not redact screen text content. DBNet [29] and CRAFT [6] detect scene text but do not address faces and were not evaluated on screen text in egocentric conditions. TrOCR [30] recognises document text but was not tested on oblique or artefact-affected screen content. Every generative anonymisation paper in Section III (from StyleGAN [14] through DeepPrivacy2 [16] and DIFP [19]) evaluates face de-identification in isolation, treating face privacy and screen/text privacy as independent problems. Wang et al. [31] measure the utility cost of resolution reduction but test neither face anonymisation nor text redaction.

This structural absence reflects what the thematic analysis identifies as the field’s foundational assumption: that PII in images is co-located with face regions. For studio portrait datasets (CelebA-HQ, FFHQ) this assumption holds. For egocentric lifelog data it does not. A CASTLE frame in which all 10 participant faces are successfully anonymised but a visible laptop screen displays an email inbox, a shared calendar, or a password prompt has not been privacy-protected: it has had one of multiple PII vectors addressed while the others are left intact. Under GDPR, biometric data constitutes a special category requiring explicit lawful basis only when processed for the purpose of uniquely identifying a natural person (Art. 9(1)), where biometric data is defined as data resulting from specific technical processing of physical or behavioural characteristics (Art. 4(14) GDPR). A partially redacted lifelog frame where facial processing enables re-identification, or where screen content discloses identifying information, may remain legally non-distributable even after complete face anonymisation.

Gap G3 is therefore stated as follows: no existing method in the reviewed corpus provides an integrated pipeline that jointly detects and redacts faces, screen-visible content, and scene text in a single egocentric frame under the imaging constraints of 4K resolution, oblique viewing angles, WebP compression, and an efficient on-device compute budget. Addressing this gap (even at a first-integration level) is the core scope of RQ3 and the primary engineering contribution of this practicum. Table V summarises all five Theme 3 papers across the evaluation dimensions most relevant to CASTLE deployment, making the evidence for Gap G3 explicit and audit-traceable.

V. PRIVACY EVALUATION FRAMEWORKS AND UTILITY METRICS

The house-standard evaluation protocol defined in this section is used retroactively as the benchmark for all methods reviewed in Sections II–IV; table entries marked N/R are therefore omissions relative to this project protocol, not necessarily defects in the original papers.

The metric inconsistency documented in Table IV (where no two anonymisation papers in the corpus report the same complete metric set) is not an accident of individual paper choices. It reflects the absence of a field-wide consensus on what evaluation constitutes evidence of privacy and what constitutes evidence of utility. Papers report whichever metric favours their method: a GAN paper may report SSIM because its perceptual quality is strong while omitting re-identification rate because its privacy protection is weak;

TABLE V
SCREEN & TEXT DETECTION METHODS , COMPARATIVE SUMMARY

Citation	Architecture	Dataset	Key Metric (Source)	Ego?	Key Limitations (CASTLE)	Assessment
Li et al. 2026 MV-VLM [28]	VLM (Llama2-7B + multi-view)	Custom STT dataset (30 children, 1,800 images)	Overall accuracy = 0.9561 (Tab. II); screen accuracy 95.5%, sensitivity 93.67% (Tab. III) [28]; per-class F1 for TV, smartphone, computer reported in Tab. II vs YOLOv11	YES	No CASTLE eval; STT-only dataset; same-stream temporal views (no cross-camera); oblique/WebP untested; no text redaction layer; multi-view conditional on stream timestamps; 2×H100 training; 0.3 s/group inference	H100 training infeasible on-device; inference-only path; no text redaction; no fine-tune; single-view fallback if no sync metadata
Wang et al. 2023 Privacy vs Activity [31]	Resolution-based privacy (CHI egocentric study)	Custom CHI egocentric	Activity recognition acc: ViT 81.0%–94.6%, ResNet50 63.9%–88.8% across resolution range 15×15 to 240×240 (Tab. 2)	YES	Resolution-reduction only tested; no generative anon; no text redaction; subjective privacy scores; small dataset	Establishes utility cost principle only; not deployable as detection method
Liao et al. 2020 DBNet [29]	Segmentation (ResNet + diff. binarisation)	MSRA-TD500 + ICDAR 2015 + Total-Text	F = 82.8%, 62 FPS (MSRA-TD500, ResNet-18; Tab. 6); ICDAR 2015: Hmean=82.3%, P=86.8%, R=78.4% (ResNet-18, 736; Tab. 5); Total-Text: F=82.8% (Tab. 3)	NO	2020 method; no oblique screen test; no WebP test; no egocentric eval; natural scene text only; FPS from MSRA-TD500 not 4K WebP	Throughput scalability baseline (62 FPS MSRA-TD500); CASTLE transfer unverified
Baek et al. 2019 CRAFT [6]	Character-region + affinity (weakly supervised)	ICDAR 2013/15/17 + MSRA-TD500 + TotalText + CTW-1500	ICDAR 2015: F=86.9% (Tab. 1); TotalText: F=83.6% (Tab. 2); CTW-1500: F=83.5% (Tab. 2)	NO	2019 foundational; slower than DBNet; no egocentric eval; TotalText drop vs ICDAR shows domain sensitivity; WebP untested	Accuracy baseline (F=86.9% ICDAR); domain shift risk on oblique screens
Li et al. 2022 TrOCR [30]	Transformer (ViT encoder + LM decoder)	IAM + SROIE	IAM handwritten: CER 2.89% (Tab. 4); SROIE printed: word-level F1 96.58% (Tab. 3)	NO	Clean frontal docs; no oblique angle; no Moiré test; no WebP artefact test; egocentric screen transfer untested	OCR baseline for screen text content; oblique/WebP performance unknown

Note: No paper reports detection or recognition performance on egocentric screen text at oblique angles, under WebP artefacts, or in the presence of Moiré interference. F-measure values from ICDAR and TotalText are natural-scene benchmarks and are not transferable to CASTLE’s screen capture conditions without re-evaluation. The 3 pp. CRAFT TotalText gap (86.9% → 83.6%) illustrates domain sensitivity; CASTLE’s oblique screen text is likely harder still. Gap G3 (no unified face + screen + text pipeline) is supported by all five rows of this table.

a diffusion paper may report re-ID suppression under ArcFace while omitting FID (as documented in Table IV, where only 4 of 14 Theme 2 methods report FID). This section formally defines the adversary threat model relevant to CASTLE, catalogues the re-identification attack tools available in the corpus with their specific quantitative benchmarks and limitations, synthesises the selection rationale for perceptual metrics (FID [7], LPIPS [8], SSIM [9]), and establishes downstream task-oriented utility measures. The outcome is the house-standard evaluation protocol committed to in Section III, applied uniformly across all methods regardless of what each paper originally reported.

A. Formal Adversary Threat Model for CASTLE

A threat model specifies who the attacker is, what they know, and what constitutes a successful attack. In the anonymisation literature, this is almost never stated formally. Papers typically assume an attacker who applies a single face recognition model to isolated image crops and reports success as re-identification of a known individual. This assumption is systematically inadequate for egocentric lifelog data and for CASTLE in particular, for three reasons. First, CASTLE has a closed-set identity pool of 10 active participants per recording day [4]. We expect (and will test) that this closed-set structure yields higher identity inference accuracy than open-world benchmarks, because the attacker’s prior probability per frame is 1-in-10 rather than 1-in-thousands; this expectation will be validated by applying EgoPrivacy’s demographic attack battery pre- and post-anonymisation on a balanced CASTLE subset stratified by participant (closed-set classifier

protocol, balanced per-identity sampling). Second, the 16-stream offline WebP corpus creates a cross-view attack surface: an attacker can use exocentric fixed-camera frames as clean identity references against which anonymised egocentric frames are matched [4]. Third, contextual and demographic signals can leak identity without any face processing at all, as EgoPrivacy [3] demonstrates. The threat model therefore comprises three attacker types: a closed-set face recognition attacker (primary); a cross-view re-identification attacker using multi-stream CASTLE footage (secondary, novel); and a contextual demographic inference attacker operating on scene content without face access (boundary condition, following [3]).

B. Re-ID Attack Tools: ArcFace and AdaFace

ArcFace [5] applies an additive angular margin penalty to the softmax loss function, producing highly discriminative face embeddings. On IJB-B and IJB-C using an MS1MV2-trained ResNet100, ArcFace achieves 94.2% TAR@FAR=1e-4 on IJB-B and 95.6% TAR@FAR=1e-4 on IJB-C [5], and 99.83% accuracy on LFW [5]. These figures are predicated on near-frontal, well-lit, high-resolution face crops. ArcFace applies a fixed margin penalty regardless of input image quality; on motion-blurred, partially occluded, or low-resolution face regions (conditions CASTLE frames systematically produce), the model’s discriminative power degrades without any internal compensatory mechanism.

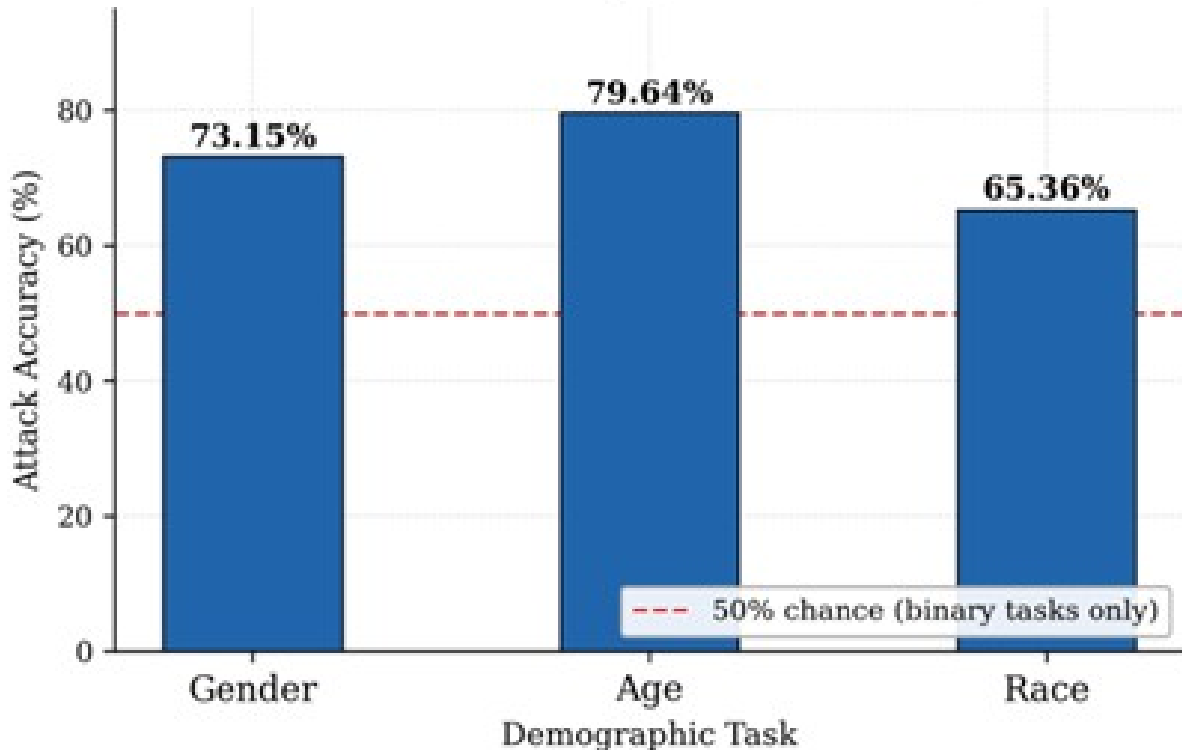
AdaFace [10] addresses this by making the margin penalty adaptive to estimated image quality at inference time: low-quality inputs receive reduced margin emphasis, preventing the model from being anchored to unreliable features. The most direct head-to-head comparison is found in [10], where both methods are evaluated on identical conditions (MS1MV2 training data, ResNet100 backbone, TAR@FAR=1e-4 threshold): ArcFace achieves 96.03% on IJB-C, while AdaFace achieves 96.89% on IJB-C [10]. The +0.86 pp. gap on this mixed-quality benchmark widens substantially on specifically low-quality data: on the IJB-S Surveillance-to-Single protocol (the closest published analog to CASTLE’s motion-blurred wearable-camera face crops), AdaFace achieves Rank-1 accuracy of 65.26% versus ArcFace’s 57.35%, an absolute improvement of +7.91 pp. under identical training and evaluation conditions [10]. AdaFace also achieves 99.82% on LFW [10]. These two comparisons together (a marginal IJB-C gap that grows to +7.91 pp. on low-quality inputs) provide the clearest evidence that AdaFace’s quality-adaptive design specifically benefits degraded egocentric captures of the type CASTLE produces.

The selection rationale for the house standard follows directly: ArcFace [5] is retained as the secondary attacker for cross-study comparability with prior anonymisation papers (all Theme 2 papers use ArcFace as their attacker), while AdaFace [10] is designated the primary attacker because its quality-adaptive margin more faithfully models a strong attacker on CASTLE’s degraded inputs. The IJB-S caveat is important: IJB-S uses fixed surveillance camera blur, whereas CASTLE produces wearable-camera motion blur (different degradation patterns). The IJB-S gap of +7.91 pp. therefore provides a directional lower bound on AdaFace’s advantage, not a precise CASTLE-specific estimate; the latter will only emerge from CASTLE re-ID evaluation itself.

C. EgoPrivacy: The Broader Threat Beyond Face Recognition

Li et al. [3] introduce EgoPrivacy, a seven-task threat model for egocentric video structured across three privacy dimensions: individual identity (re-identification), demographic attributes (gender, age, race), and situational context (activity, scene). The critical finding is that zero-shot foundation models achieve 73.15% gender attack accuracy, 79.64% age attack accuracy, and 65.36% race attack accuracy at peak across models (Tab. 2 of [3]) on Ego-Exo4D (in-distribution) and Charades-Ego (out-of-distribution) egocentric footage without any face detection, operating solely on scene context, posture, and object interactions [3]. Chance accuracy for a binary demographic classifier is 50%; the EgoPrivacy results sit 15–30 percentage points above chance across all three tasks under zero-shot conditions. These results confirm the fundamental inadequacy of face-only re-ID evaluation: an anonymisation pipeline that reduces ArcFace re-ID from 89% to 3% [14] while leaving demographic and contextual signals intact has not meaningfully protected the privacy of a CASTLE participant under the EgoPrivacy threat model.

Figure 3. EgoPrivacy Zero-Shot Demographic Attack Accuracy (No Face Access)



† 50% baseline applies to binary tasks (Gender) only; Age and Race are multi-class with non-uniform baselines.

Fig. 3. EgoPrivacy zero-shot demographic attack accuracy without any face access. Source: [3] Table 2. The 50% chance reference line applies to Gender (binary) only; Age and Race are multi-class tasks. Closed-set CASTLE attacks (10 active participants per day) may exceed these open-set results. Cross-reference Section V, Gap G1, Tab. VII, RQ4.

EgoPrivacy [3] was evaluated on Ego-Exo4D (in-distribution) and Charades-Ego (out-of-distribution test set), not on CASTLE’s closed-set per-day pool of 10 active participants drawn from 11 unique identifiers. We hypothesise that CASTLE’s closed-set structure will yield higher demographic inference accuracy than EgoPrivacy reports, given the higher prior probability per frame in a 10-person pool versus thousands of Ego4D participants; this will be tested by applying EgoPrivacy’s demographic attack battery on a per-participant balanced CASTLE subset under both closed-set classifier and open-set verification protocols. Applying this seven-task evaluation framework pre- and post-anonymisation on a CASTLE subset would, to our knowledge within the reviewed egocentric anonymisation literature, constitute a more comprehensive privacy evaluation than any single prior lifelog anonymisation study.

D. Perceptual Utility Metrics: FID, LPIPS, and SSIM

Fréchet Inception Distance [7] measures the distance between the Inception-V3 feature distributions of a real image set and a generated image set; FID = 0 indicates identical distributions. [7] is cited here for the metric definition and computation only. Observed anonymisation FID values come from the locked corpus: DeepPrivacy2 [16] reports FID = 0.56 on the FDF128 face dataset; DIFP [19] FID = N/R (not measured; paper uses PSNR/SSIM for recovery quality only); Kung et al. [21] FID = N/R (not reported; paper uses re-identification rate, face shape distance, and IQA score). These values reflect the GAN-to-diffusion quality range documented in Section III, but all verified values are evaluated on portrait datasets and are not transferable to CASTLE’s egocentric frames without re-evaluation. For this project, FID will be computed

on a representative 10,000-sample face-crop subset drawn from the 416,542-frame offline WebP corpus due to Inception-V3 compute cost on practical on-device compute constraints. Critically, FID is computed against WebP-compressed originals, not lossless references; this must be documented explicitly in the methodology chapter to prevent mis-attribution of WebP compression loss as anonymisation degradation. The unanonymised CASTLE WebP FID baseline is therefore computed first, before any anonymised FID value is reported.

Learned Perceptual Image Patch Similarity [8] measures perceptual dissimilarity between image patches using deep VGG features calibrated against human perceptual judgements; LPIPS = 0 indicates perceptual identity. [8] is cited for the metric definition only. Within the locked corpus, Diff-Privacy [20] reports using LPIPS but no exact value is available in the accessible abstract (N/R); no other locked corpus paper provides an audit-traceable LPIPS range for anonymisation outputs. LPIPS is nonetheless selected as the most sensitive perceptual metric for detecting generative anonymisation artefacts (blending boundaries, texture inconsistencies, and over-smoothing) that FID and SSIM may miss. The VGG backbone was trained on natural images, not egocentric lifelog content; the perceptual calibration may not perfectly reflect what a human viewer in this context considers natural, and this must be acknowledged as a metric limitation.

Structural Similarity Index Measure [9] evaluates luminance, contrast, and structural correlation; SSIM = 1.0 indicates perfect structural identity. [9] is cited for the metric definition only. Within the locked corpus: CPP-DeID [15] reports SSIM = 0.74 (abstract-level only; full IVC paper behind paywall); ReFaceX [23] reports post-reversal SSIM recovery of 0.9378 [23] (verified). These corpus values establish an indicative range of approximately 0.74–0.94 for published anonymisation methods on portrait datasets. For this project, $\text{SSIM} \geq 0.75$ is adopted as a soft quality gate (a project-defined design choice, not a field-wide standard), meaning frames below this threshold will be flagged for manual inspection rather than hard-rejected. SSIM will be computed against WebP-compressed originals. Its known limitation (insensitivity to perceptual realism) motivates using SSIM alongside LPIPS rather than as a substitute.

E. Downstream Task Utility and Validation Frameworks

Perceptual metrics assess image quality in isolation; they do not measure whether anonymised frames remain useful for the downstream tasks CASTLE was designed to support: Event Search (mAP), Object Search (Recall@k), and Visual Question Answering (F1) [4]. Stenger et al. [32] systematically evaluate how anonymisation methods affect downstream computer vision task accuracy, reporting task-accuracy degradation as the primary measure [32]. While [32] does not use egocentric data, it establishes the methodological principle that anonymisation must be evaluated on task output, not only image quality. For this project, CASTLE’s official Event Search mAP serves as the primary downstream utility gate: anonymisation is viable only if post-anonymisation mAP degradation remains within a pre-specified threshold defined in the evaluation protocol.

EPIC-Kitchens-100 [33] provides the closest published egocentric action recognition benchmark, with TSN and stronger baselines achieving verb top-1 accuracies in the 60–70% range on the public leaderboard [33]. EPIC-Kitchens serves as a downstream utility proxy when CASTLE-specific event search evaluation is infeasible: if anonymised frames preserve action recognition accuracy at near-baseline levels, egocentric utility is considered maintained. The kitchen-vs-collaborative-office domain gap must be acknowledged as a proxy limitation. Khorzooghi and Nilizadeh [14] provide an attribute preservation utility framework (age mean absolute difference of 5 and gender match rate of 98% for StyleGAN 0-8 de-identification, Tab. 2 of [14]), which serves as the baseline for evaluating whether anonymisation-induced attribute distortion is within socially interpretable bounds. MAP2V [34] provides a systematic validation framework applied to this project’s CASTLE results as a post-hoc methodological rigour check [34].

VI. GAPS AND RESEARCH QUESTIONS

The five preceding sections have produced not isolated critiques but a converging argument. Three independent lines of evidence (the degradation-blindness of fixed-strategy anonymisation (Section III),

TABLE VI
PRIVACY EVALUATION FRAMEWORKS AND ATTACKER MODELS

Citation	Method Type	Dataset	Key Metric (Source)	Attack Mechanism	CASTLE Limitation	CASTLE Protocol
Li et al. 2025 EgoPrivacy [3]	7-task threat (demographic + ID + situational)	Ego-Exo4D (in-dist.) + Charades-Ego (OOD test)	Gender: 73.15%; Age: 79.64%; Race: 65.36% (peak across models, zero-shot, no face; Tab. 2 of [3])	Context-only; no face needed	Not on CASTLE closed-set; closed-set hypothesised stronger	Use 7-task battery pre/post-anon CASTLE subset; closed-set + open-set protocols
Zhang et al. 2024 MAP2V [34]	Systematic validation framework	Multiple anon datasets	Qualitative validation score (Sec. 4 Validation Framework)	Framework (not an attacker)	Method- dependent; no standalone privacy metric	Post-hoc validation check on CASTLE results
Deng et al. 2019 ArcFace [5]	Face recog. (fixed angular margin, $m=0.5$)	MS1MV2 + LFW + IJB-B/C	LFW: 99.83% (Tab. 4); IJB-B: 94.2% @ $1e-4$; IJB-C: 95.6% @ $1e-4$ (Tab. 7, own paper); IJB-C: 96.03% @ $1e-4$ (AdaFace Tab. 3(a) identical conditions)	Closed-set face re-ID; cosine sim; fixed margin	Fixed margin degrades on blur; no quality adapt.	SECONDARY attacker (prior-work comparability)
Stenger et al. 2025 [32]	Downstream task impact eval.	Multiple non-ego CV datasets	CV accuracy drop (method-dependent) (Sec. 4 Evaluation)	Downstream task accuracy as failure signal	Non-egocentric; CASTLE- specific impact needs separate measure	Apply methodology to CASTLE Event Search mAP gate
Kim et al. 2022 AdaFace [10]	Face recog. (quality-adaptive margin, $m=0.4$)	MS1MV2 R100	LFW: 99.82% (Tab. 3(a)); IJB-B: 95.67% @ $1e-4$; IJB-C: 96.89% @ $1e-4$ (Tab. 3(a), MS1MV2) vs ArcFace 96.03% same conditions; IJB-S S-to-Single Rank-1: 65.26% vs ArcFace: 57.35% (Tab. 3(b), MS1MV2)	Quality- adaptive margin; feature norm as quality proxy	IJB-S = closest analog to CASTLE blur; fixed surv. camera not equal to wearable motion blur	PRIMARY attacker; +0.86pp IJB-C (identical cond.); +7.91pp IJB-S low-quality; quality-adaptive

FAR THRESHOLD NOTE: Both ArcFace [5] and AdaFace [10] use $\text{TAR@FAR}=1e-4$ ($= 0.01\%$). The thresholds are consistent; a direct IJB-C comparison is valid. For head-to-head comparison, use AdaFace Tab. 3(a) where both methods are evaluated under identical conditions: ArcFace IJB-C = 96.03%, AdaFace IJB-C = 96.89% @ $\text{FAR}=1e-4$. Note: ArcFace’s own Tab. 7 reports IJB-C = 95.6% under its own protocol (minor re-implementation difference). PRIMARY attacker = AdaFace [10]; SECONDARY = ArcFace [5].

the structural face-only assumption exposed by EgoPrivacy (Section V), and the absence of a unified multimodal pipeline (Section IV)) each independently problematise the state of the art and together define the research space this project occupies.

A. Evidence-Backed Gap Analysis

Gap G1 (Absence of quality-aware adaptive method selection) (PRIMARY GAP): Within the locked 34-paper corpus, no reviewed method explicitly performs input-quality-aware strategy selection, that is, no method chooses between GAN, diffusion, or perturbation approaches based on measurable face quality indicators such as resolution, blur level, or occlusion; most apply a fixed transformation rule per detected face regardless of input condition. This is not merely a design simplification; it is an assumption that the privacy-utility trade-off is stable across all input conditions. The assumption collapses under CASTLE’s imaging constraints: AdaFace’s +7.91 pp. advantage over ArcFace on IJB-S low-quality inputs [10] demonstrates that the attacker’s success is input-quality-dependent, yet within the reviewed corpus no anonymisation method calibrates its strategy to this variability. G1 is the central research problem, directly defining the project’s primary contribution: an input-quality-aware selection mechanism that dynamically assigns an anonymisation strategy per face based on measurable quality indicators within a 4K egocentric frame. The router quality signals are defined as estimated blur level via Laplacian variance, motivated by egocentric motion degradation [27]; detected face size in pixels, motivated by scale variation in de-identification surveys [2]; estimated occlusion ratio, motivated by motion-consistent privacy obfuscation constraints [26]; and WebP artefact severity, motivated by CASTLE’s released still-frame format [4]. Their

TABLE VII
UTILITY METRICS , DEFINITION, RANGE, AND CASTLE APPLICATION

Citation	Metric Definition	Direction & Range	Value Source (definition vs corpus)	CASTLE Application	CASTLE-Specific Limitation	Selection Rationale
Heusel et al. 2017 FID [7]	Inception-V3 feature dist. between real and generated sets	Lower = better; 0 = identical distributions	Definition: cite [7]. Corpus ranges (verified only): DeepPrivacy2 [16] FID=0.56 (FDF128); DIFP [19] FID=N/R; Kung [21] FID=N/R. (All portrait data)	10k-face CASTLE subset; WebP baseline; state subset method	Compute-heavy; sample-sensitive; corpus values on portraits not ego	Distribution realism; broadest corpus coverage; verified range: DeepPrivacy2 = 0.56 (only verified corpus value)
Zhang et al. 2018 LPIPS [8]	VGG deep feature patch similarity vs human judgements	Lower = better; 0 = identical	Definition: cite [8]. Corpus: Diff-Privacy [20] uses LPIPS but exact value N/R (paywall abstract). No other locked corpus value.	Per-face crop; WebP reference; document VGG naturalness limit	VGG on natural images; WebP ref. lossy; no corpus range available	Sensitive to GAN artefacts; use alongside SSIM
Wang et al. 2004 SSIM [9]	Luminance $\times$ contrast $\times$ structure correlation	Higher = better; max 1.0	Definition: cite [9]. Corpus ranges (corrected): ReFaceX [23] 0.9378 (Tab. 6, recovered image quality on CelebA-HQ, verified). CPP-DeID [15] 0.74 (abstract-level only; paywall). Range: $\sim$ 0.74 (directional) to 0.9378 (verified).	CPU-only; per face crop; WebP reference; soft gate ≥ 0.75 (project choice, not field std.)	Insensitive to perceptual realism; WebP ref. lossy; corpus values on portraits	CPU-fast; corpus range $\sim$ 0.74–0.94; use with LPIPS
Damen et al. 2022 EPIC-Kitchens [33]	Egocentric action recog. (downstream proxy)	Higher = better; top-1 verb accuracy	TSN and stronger baselines: verb top-1 in 60–70% range (public leaderboard)	Post-anon frames; egocentric task proxy; CASTLE Event Search mAP = primary	Kitchen domain; CASTLE = collab. indoor; domain gap to acknowledge	Ego downstream proxy; use when CASTLE benchmark infeasible
Khorzooghi & Nilizadeh 2023 Utility Framework [14] [†]	Attribute preservation (age mean diff + gender match rate)	Higher = better (attribute acc.)	Age mean diff: 5; Gender match: 98% (StyleGAN 0-8, Tab. 2, Face++, FFHQ)	Attribute eval. on CASTLE subset post-anon	FFHQ only; no motion-blur; portrait cond.	Socially interpretable utility check; attribute preservation

NOTE: Metric papers [7][8][9] cited for DEFINITIONS ONLY. Observed value ranges derived from locked corpus anonymisation papers [15][16][23][21]. All SSIM and LPIPS computed against WebP-compressed originals; baseline includes WebP loss; document explicitly in methodology chapter. SSIM ≥ 0.75 = project-defined soft gate, not field standard. N/R = Not Reported (specific value from full journal table not accessible). DIFP [19] FID = N/R. AdaFace [10] PRIMARY + ArcFace [5] SECONDARY for re-ID; LPIPS [8] + SSIM [9] (soft gate ≥ 0.75 , project-defined) + subset FID [7] for perceptual quality; CASTLE Event Search mAP (primary) / EPIC-Kitchens [33] (proxy) for downstream utility; EgoPrivacy 7-task battery [3] for broader demographic threat; MAP2V [34] post-hoc validation.

thresholds are derived empirically through sensitivity analysis on a dedicated 200-frame calibration set distinct from the development and evaluation subsets.

Gap G2 (Absence of cross-view consistency evaluation) (SECONDARY): CASTLE’s multi-stream WebP corpus creates a cross-view identity leakage surface not modelled in any reviewed anonymisation evaluation [4]. An identity anonymised in an egocentric frame may remain visually consistent and re-identifiable across fixed exocentric views. The project can evaluate this gap using CASTLE’s filename-derived temporal alignment: filenames follow the ‘HHNNNN’ pattern, where ‘HH’ denotes hour of day and ‘NNNN’ is a frame counter sampled at 5-second intervals, giving the validated offset rule $t = (NNNN - 1) \times 5$ seconds. The remaining limitation is therefore alignment granularity and offline still-frame access, not total infeasibility.

Gap G3 (Absence of a unified face-plus-screen privacy pipeline) (ENGINEERING GAP): As established in Section IV, within the reviewed corpus not one paper simultaneously addresses face anonymisation and screen/text redaction in the same egocentric frame. Under EU GDPR, facial imagery can constitute biometric data when processed for the purpose of uniquely identifying a natural person (Art. 4(14); Art. 9(1), Regulation (EU) 2016/679); screen-visible content may independently constitute personal data. Face-only processing may therefore be insufficient for compliant data sharing under a strict privacy threat model. G3 defines the project’s secondary engineering contribution: a modular multimodal pipeline integrating

face detection, screen detection, and OCR-based text redaction on CASTLE frames within an efficient on-device compute budget.

B. Derived Research Questions

RQ1 addresses the prerequisite detection problem: To what extent does YOLOv9 (fine-tuned for a dedicated face class on an annotated CASTLE subset) outperform RetinaFace in detecting faces under CASTLE’s egocentric degradation conditions (motion blur, non-frontal angles, WebP compression), when both models are fine-tuned on identical data and evaluated under identical metrics (AP@0.5:0.95, recall at varied IoU thresholds)? This question is logically prior to RQ2: adaptive method selection requires reliable detection, and it is motivated by Section II’s evidence that benchmark detection performance is systematically decoupled from egocentric robustness [11], [12].

RQ2 addresses the face anonymisation evaluation gap: How do GAN-based, diffusion-based, and standard perturbation (blur and pixelation) face anonymisation methods compare with respect to privacy-utility trade-offs (specifically re-identification resistance and perceptual quality preservation) when applied to egocentric lifelog imagery under a standardized evaluation protocol? This question is motivated by the metric inconsistency exposed in Section V, where no two papers report the same complete metric set, and it establishes the comparative baseline for evaluating anonymisation effectiveness. Pairwise method comparisons will use paired t-tests for SSIM and LPIPS, McNemar tests for detector comparisons, and Wilcoxon signed-rank tests for re-identification comparisons, with 95% confidence intervals reported alongside means.

RQ3 addresses G3: Can a unified multimodal pipeline (YOLOv9 face detection, MV-VLM screen detection with single-view fallback, TrOCR text recognition, and adaptive face anonymisation) achieve comprehensive PII removal from CASTLE frames without degrading CASTLE’s image-based retrieval benchmark (or equivalent image-only proxy) by more than 10% (provisional threshold, to be refined from the unanonymised CASTLE baseline)? This question is motivated by Section IV’s structural finding that screen/text and face privacy have never been jointly evaluated on lifelog data [6], [29], [30].

RQ4 addresses G1 directly: Does an adaptive selection mechanism (choosing between GAN, diffusion, and perturbation strategies based on detected face quality metrics like estimated blur, face size, occlusion, and WebP severity) achieve a superior privacy-utility trade-off compared to any single fixed-strategy method, evaluated on CASTLE 2024 using the house-standard protocol (AdaFace [10] + ArcFace [5] + EgoPrivacy tasks [3] + LPIPS [8] + SSIM [9] + FID subset [7] + CASTLE retrieval benchmark or image-only proxy)? This is the central empirical question of the project and the direct experimental test of Gap G1. RQ4 also incorporates the cross-view privacy analysis (G2) enabled by temporally aligned egocentric and exocentric streams to measure identity leakage consistency across different camera viewpoints.

VII. CONCLUSION

A. Synthesis of Key Analytical Findings

This review has examined 34 papers across four thematic areas (face detection in egocentric data, generative face anonymisation, multimodal screen and text privacy, and privacy evaluation including utility metrics), anchored by the CASTLE 2024 dataset, to characterise the state of the art for privacy-preserving anonymisation of high-resolution lifelog data. Across all themes, a consistent pattern emerges: the field has made substantial technical progress on individual sub-problems while systematically failing to integrate them under realistic egocentric evaluation conditions.

Theme 1 found that among the strongest detectors in this review (YOLOv9 [12] at mAP=53.0% on COCO and RetinaFace [11] at AP=91.4% on WIDER FACE Hard), neither has been evaluated under CASTLE’s concurrent conditions of 4K resolution, wearable-camera motion blur, non-frontal angles, and WebP compression. Theme 2 documented a GAN-to-diffusion-to-reversible quality progression. StyleGAN [14] achieves identification attack accuracy (FaceNet+SVM, m1) of 11.7% at the privacy-optimal StyleGAN 0-3 level (Tab. 3), and DeepPrivacy2 [16] achieves FID = 0.56. Reversible methods

TABLE VIII
GAP-TO-RQ MAPPING , EVIDENCE TRACEABILITY

Gap	Description	Evidence Trail (Section + Citations)	Maps to RQ	Contribution Type
G1 (PRIMARY)	No quality-aware adaptive strategy selection in any reviewed method (within 33-paper corpus)	Section III: fixed-strategy assumption; AdaFace +7.91pp IJB-S [10]; all Theme 2 papers [2], [14]–[21], [23]–[27]	RQ4 (adaptive vs fixed on CASTLE; house-standard protocol [3], [5], [7]–[10])	Central research problem; all other RQs depend on G1
G2 (CONDITIONAL)	No cross-view anonymisation consistency evaluation in reviewed literature	Section II / Section V: CASTLE 15 time-aligned streams [4]; no timestamp metadata in project WebP set	RQ4 cross-view sub-task if sync metadata obtained	Conditional on metadata; document as limitation if absent
G3	No unified face + screen/text pipeline in reviewed corpus on egocentric lifelog data	Section IV: structural separation confirmed across all reviewed papers [6], [28]–[30]; EU GDPR Art. 9(1) + Art. 4(14) legal framing	RQ3 (multimodal pipeline vs image-based retrieval benchmark (proxy))	Engineering contribution; first integration attempt
Eval. Gap	No anonymisation paper applies EgoPrivacy 7-task model as post-anon evaluation	Section V: EgoPrivacy [3] 73.15/79.64/65.36% demographic attack without face access, peak across models (Tab. 2 of [3]); no Theme 2 paper uses this model	RQ2 (standardised re-ID and perceptual quality evaluation battery pre/post-anonymisation)	Evaluation methodology gap; strengthens novelty claim
Pre-cond.	Detection performance decoupled from egocentric robustness in reviewed corpus	Section II: YOLOv9 [12] COCO mAP vs RetinaFace [11] WIDER FACE; different benchmarks, no CASTLE evaluation	RQ1 (fair detection benchmark on CASTLE; experimental equalizer (face class fine-tune))	Logical prerequisite for RQ2; not a gap but enabling question

NOTE: Each RQ derives from a specific gap; each gap is grounded in specific section findings and corpus citations. G1 is the PRIMARY gap defining the central research problem and mapping to RQ4. G2 (conditional) maps to RQ4 if metadata obtained. G3 maps to RQ3. RQ1 is a logical prerequisite. RQ2 addresses an evaluation methodology gap and anonymisation comparison. Logical derivation chain: Evidence in Sections II–V → Gaps G1/G2/G3 + Evaluation Gap → RQ1 (prerequisite detection) → RQ2 (anonymisation comparison and evaluation standardisation) → RQ3 (engineering: multimodal pipeline vs G3) → RQ4 (adaptive selection and cross-view evaluation). This chain makes the evidential parent of each RQ explicit.

then show stronger identity suppression: ReFaceX [23] reports post-anonymisation cosine similarity of 0.009 / 0.008 / 0.017 (FaceNet/ArcFace/AdaFace on LFW, Tab. 4), while NDSS key-driven anonymisation [24] reports an unsuccessful authentication rate (UAR) of 0.962. These are strong results, but they are obtained predominantly on controlled portrait datasets (CelebA-HQ, FFHQ) under conditions that do not transfer to CASTLE’s egocentric frames. Theme 3 established that screen and text detection methods [6], [28]–[30] and face anonymisation methods [2], [14]–[27] have never been evaluated jointly in the same pipeline on the same frame (a structural separation that breaks down for any dataset where both modalities co-occur). The privacy evaluation theme exposed the most fundamental contradiction: EgoPrivacy [3] demonstrates that zero-shot foundation models achieve demographic attack accuracies of up to 79.64% (Gender: 73.15%, Age: 79.64%, Race: 65.36% at peak across models) (Tab. 2 of [3]) on egocentric video without any face access, yet every Theme 2 paper in this review evaluates privacy success exclusively at the face level.

B. Strengths and Weaknesses of the Existing Literature

The literature’s strengths are real: reversible anonymisation methods [23], [24] directly address the research-access tension inherent in lifelog datasets; quality-adaptive face recognition [10] provides a principled attacker model for degraded imagery; and EgoPrivacy [3] represents a meaningful step toward comprehensive, multi-dimensional privacy evaluation. Metric standardisation is advancing: ArcFace [5], FID [7], LPIPS [8], and SSIM [9] are increasingly consistently reported. The weaknesses are equally clear. First, the literature’s predominant reliance on CelebA-HQ and FFHQ introduces a systematic studio-conditions bias: performance figures for most Theme 2 methods are contingent on clean, aligned, high-resolution inputs that CASTLE’s wearable-camera frames do not provide. Second, metric inconsistency persists (no two Theme 2 papers report the same complete metric set), making cross-method comparison from reported values alone impossible without standardised re-evaluation. Third, and most critically, no reviewed paper

addresses the joint privacy threat arising when faces, screens, and contextual signals co-occur in a single egocentric frame, the precise condition CASTLE represents.

C. Ethical Considerations

The ethical stakes of this work are grounded in three concrete obligations. First, under EU GDPR (Regulation (EU) 2016/679), facial imagery processed for the purpose of uniquely identifying a natural person constitutes special-category biometric data (Art. 4(14); Art. 9(1)), requiring a lawful basis for any processing or sharing. CASTLE’s ten participants provided full written informed consent and reviewed their own footage prior to release [4]; the dataset is distributed under a CC BY-NC-SA 4.0 licence (non-commercial research use only), placing a legal obligation on this project to implement, not merely discuss, privacy protection. Second, the CC BY-NC-SA licence creates a specific tension: authorised researchers need access to identifiable faces for annotation and ground-truth validation, while any broader distribution requires those faces to be protected. This tension motivates the inclusion of reversible anonymisation methods (ReFaceX [23], NDSS key-driven [24]) as first-class candidates: key-controlled reversibility enables controlled researcher access without distributing identifiable data. Third, the application of automated re-identification models to a real dataset of ten named, consenting individuals requires careful scoping: attack evaluation is conducted solely to measure anonymisation effectiveness, not to produce or store re-identified outputs, and access to CASTLE requires DCU ethical approval, which has been obtained for this project. In line with the GDPR data minimisation principle (Art. 5(1)(c), Regulation EU 2016/679), re-identification models are applied to anonymised outputs only; no identifiable face embeddings are retained or transmitted beyond the evaluation environment.

D. Lead-in to Methodology

The gaps formalised in Section VI (the absence of quality-aware adaptive method selection (G1), the unaddressed cross-view identity surface in multi-stream data (G2), and the absence of a unified multimodal privacy pipeline (G3)) converge on a single omission: to the best of our knowledge within the reviewed literature, no prior work has built and evaluated an integrated anonymisation framework on a real-world lifelog dataset combining a closed identity pool, multi-stream capture, and standardised downstream benchmarks. CASTLE 2024 [4] uniquely satisfies all three conditions. The methodology that follows is constructed directly from the constraints surfaced by this review, all within an efficient on-device compute budget: an AdaFace-primary [10] threat model calibrated to degraded inputs; a YOLOv9 [12] detection stage fine-tuned on a dedicated face class; quality-conditioned strategy selection across the GAN-diffusion-reversible spectrum; MV-VLM [28] screen detection with single-view fallback; TrOCR [30] text redaction; and a house-standard evaluation protocol spanning EgoPrivacy’s seven-task battery [3], perceptual quality metrics [7]–[9], and image-based retrieval benchmarks derived from CASTLE’s official evaluation suite [4]. The application artefact is a downstream translational output built on the validated research pipeline, not the primary academic contribution. Subset labels are accepted through model-assisted pre-labelling plus targeted human validation with provenance tracking; where validation is insufficient, labels are reported as operational proxies rather than ground truth. The study is designed to support thesis reporting and research-paper-quality comparative evidence, with statistical testing and limitation disclosure treated as mandatory. Each design decision traces directly to a finding or conflict documented in this review.

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